

0.1 Solution

Write $P(x) = x^7 + a_6x^6 + a_5x^5 + \cdots + a_1x + a_0$, so the coefficients are $a_7 = 1, a_6, \dots, a_0$ (eight in total). The conditions say every $|a_i|$ is an **odd** integer with $|a_i| < 2025^2$, and every $|a_i|$ is a **root** of P .

Since each $|a_i|$ is odd it is nonzero, so every coefficient is a nonzero odd integer and every $|a_i|$ is a **positive odd integer that is a root of P** . Let S be the set of distinct values $|a_i|$. The leading coefficient gives $1 \in S$.

Step 1: $|S| \leq 2$. Write $S = \{r_1 < r_2 < \cdots < r_k\}$ with $r_1 = 1$. Each r_i is an integer root, so

$$P(x) = \left(\prod_{i=1}^k (x - r_i) \right) Q(x), \quad Q \in \mathbb{Z}[x] \text{ monic of degree } 7 - k.$$

Then $a_0 = P(0) = (-1)^k (\prod_i r_i) Q(0)$. Since a_0 is odd, $a_0 \neq 0$, so $Q(0) \neq 0$ and $|Q(0)| \geq 1$. Hence

$$|a_0| \geq \prod_{i=1}^k r_i \quad \text{while} \quad |a_0| \leq \max S = r_k,$$

giving $\prod_{i=1}^{k-1} r_i \leq 1$. As the r_i are distinct positive integers with $r_1 = 1$, this forces $k \leq 2$.

Step 2: case $|S| = 1$. All coefficients are ± 1 and we need 1 to be a root: $P(1) = 1 + \sum_{i=0}^6 a_i = 0$. With seven terms $a_i \in \{\pm 1\}$, this means exactly three equal $+1$. There are $\binom{7}{3} = 35$ such polynomials.

Step 3: case $|S| = 2$. Here $S = \{1, m\}$ with m odd, $3 \leq m < 2025^2$, and every coefficient lies in $\{\pm 1, \pm m\}$. Write $a_i = \varepsilon_i \mu_i$ with $\varepsilon_i = \pm 1$, $\mu_i \in \{1, m\}$. Since $\mu_i m^i \in \{m^i, m^{i+1}\}$,

$$P(m) = \sum_{j=0}^7 c_j m^j, \quad c_j = \begin{cases} \varepsilon_0 [\mu_0=1], & j=0, \\ \varepsilon_j [\mu_j=1] + \varepsilon_{j-1} [\mu_{j-1}=m], & 1 \leq j \leq 6, \\ 1 + \varepsilon_6 [\mu_6=m], & j=7, \end{cases}$$

and $|c_j| \leq 2$. If some $c_j \neq 0$, taking the largest such index J gives $|c_J m^J| \geq m^J$ but $|\sum_{j < J} c_j m^j| \leq 2 \frac{m^J - 1}{m - 1} < m^J$ (using $m \geq 3$), contradicting $P(m) = 0$. Hence **all** $c_j = 0$.

Let $u_i = [\mu_i = m]$. Then:

- $c_0 = 0 \Rightarrow u_0 = 1$; $c_7 = 0 \Rightarrow u_6 = 1$, $\varepsilon_6 = -1$.
- For $1 \leq j \leq 6$, $c_j = (1 - u_j)\varepsilon_j + u_{j-1}\varepsilon_{j-1} = 0$. If $u_{j-1} = u_j = 0$ then $c_j = \varepsilon_j \neq 0$; if both $= 1$ then $c_j = \varepsilon_{j-1} \neq 0$. So $u_{j-1} \neq u_j$, and when $(u_{j-1}, u_j) = (1, 0)$ we need $\varepsilon_j = -\varepsilon_{j-1}$.

The u_i must alternate; with $u_0 = 1$ this forces $u = (1, 0, 1, 0, 1, 0, 1)$, consistent with $u_6 = 1$. Thus a_0, a_2, a_4, a_6 are $\pm m$ and a_1, a_3, a_5 are ± 1 , with $\varepsilon_6 = -1$, $\varepsilon_1 = -\varepsilon_0$, $\varepsilon_3 = -\varepsilon_2$, $\varepsilon_5 = -\varepsilon_4$, leaving $\varepsilon_0, \varepsilon_2, \varepsilon_4$ free.

Now impose $P(1) = 0$. Writing $T = \varepsilon_0 + \varepsilon_2 + \varepsilon_4$,

$$P(1) = 1 + m(\varepsilon_0 + \varepsilon_2 + \varepsilon_4 + \varepsilon_6) + (\varepsilon_1 + \varepsilon_3 + \varepsilon_5) = 1 + m(T - 1) - T = (T - 1)(m - 1).$$

Since $m \neq 1$, we need $T = 1$, i.e. exactly two of $\varepsilon_0, \varepsilon_2, \varepsilon_4$ equal $+1$: $\binom{3}{1} = 3$ **choices**. (E.g. for $m = 3$: $x^7 - 3x^6 + x^5 - 3x^4 - x^3 + 3x^2 - x + 3$, which has 1 and 3 as roots.)

So each valid m gives exactly 3 good polynomials.

Step 4: counting. Valid m are odd integers with $3 \leq m < 2025^2 = 4100625$, i.e. $3 \leq m \leq 4100623$. Their count is $\frac{4100623-3}{2} + 1 = 2050311$.

Total: $35 + 3 \cdot 2050311 = 35 + 6150933 = \boxed{6150968}$.

Step 1: $|S| \leq 2$. Write $S = \{r_1 < r_2 < \dots < r_k\}$ with $r_1 = 1$. Each r_i is an integer root, so

$$P(x) = \left(\prod_{i=1}^k (x - r_i) \right) Q(x), \quad Q \in \mathbb{Z}[x] \text{ monic of degree } 7 - k.$$

Then $a_0 = P(0) = (-1)^k \left(\prod_i r_i \right) Q(0)$. As a_0 is odd, $a_0 \neq 0$, so $Q(0) \neq 0$ and $|Q(0)| \geq 1$. Hence

$$\prod_{i=1}^k r_i \leq |a_0| \leq \max S = r_k,$$

so $\prod_{i=1}^{k-1} r_i \leq 1$. The r_i are distinct positive integers with $r_1 = 1$, so this forces $k \leq 2$.

Step 2: case $|S| = 1$. All coefficients are ± 1 , and we need 1 to be a root: $P(1) = 1 + \sum_{i=0}^6 a_i = 0$. With seven terms $a_i \in \{\pm 1\}$, exactly three must equal +1. This gives $\binom{7}{3} = 35$ polynomials, all clearly valid.

Step 3: case $|S| = 2$. Here $S = \{1, m\}$ with m odd, $3 \leq m < 2025^2$, and every coefficient lies in $\{\pm 1, \pm m\}$. Write $a_i = \varepsilon_i \mu_i$ with $\varepsilon_i = \pm 1$, $\mu_i \in \{1, m\}$. Since $\mu_i m^i \in \{m^i, m^{i+1}\}$, collecting powers of m gives $P(m) = \sum_{j=0}^7 c_j m^j$ where

$$c_0 = \varepsilon_0 [\mu_0=1], \quad c_j = \varepsilon_j [\mu_j=1] + \varepsilon_{j-1} [\mu_{j-1}=m] \quad (1 \leq j \leq 6), \quad c_7 = 1 + \varepsilon_6 [\mu_6=m],$$

and $|c_j| \leq 2$. If some $c_j \neq 0$, let J be the largest such index; then $|c_J m^J| \geq m^J$ while $|\sum_{j < J} c_j m^j| \leq 2 \frac{m^J - 1}{m - 1} < m^J$ (since $m \geq 3$), contradicting $P(m) = 0$. Hence all $c_j = 0$.

Let $u_i = [\mu_i = m]$. Then $c_0 = 0 \Rightarrow u_0 = 1$; and $c_7 = 0 \Rightarrow u_6 = 1$, $\varepsilon_6 = -1$. For $1 \leq j \leq 6$, $c_j = (1 - u_j)\varepsilon_j + u_{j-1}\varepsilon_{j-1} = 0$: if $u_{j-1} = u_j = 0$ then $c_j = \varepsilon_j \neq 0$, and if both equal 1 then $c_j = \varepsilon_{j-1} \neq 0$; so $u_{j-1} \neq u_j$, and when $(u_{j-1}, u_j) = (1, 0)$ we need $\varepsilon_j = -\varepsilon_{j-1}$. The u_i alternate; with $u_0 = 1$ this forces $u = (1, 0, 1, 0, 1, 0, 1)$, consistent with $u_6 = 1$. Thus a_0, a_2, a_4, a_6 are $\pm m$ and a_1, a_3, a_5 are ± 1 , with $\varepsilon_6 = -1$ and $\varepsilon_1 = -\varepsilon_0$, $\varepsilon_3 = -\varepsilon_2$, $\varepsilon_5 = -\varepsilon_4$, leaving $\varepsilon_0, \varepsilon_2, \varepsilon_4$ free.

Now impose $P(1) = 0$. With $T = \varepsilon_0 + \varepsilon_2 + \varepsilon_4$,

$$P(1) = 1 + m(\varepsilon_0 + \varepsilon_2 + \varepsilon_4 + \varepsilon_6) + (\varepsilon_1 + \varepsilon_3 + \varepsilon_5) = 1 + m(T - 1) - T = (T - 1)(m - 1).$$

Since $m \neq 1$, we need $T = 1$, i.e. exactly two of $\varepsilon_0, \varepsilon_2, \varepsilon_4$ equal +1: $\binom{3}{1} = 3$ choices. Each yields a genuine good polynomial (both 1 and m are roots, absolute values are exactly $\{1, m\}$). For example, with $m = 3$ one gets $x^7 - 3x^6 + x^5 - 3x^4 - x^3 + 3x^2 - x + 3$, having roots 1 and 3.

So each admissible m gives exactly 3 good polynomials.

Step 4: counting. Admissible m are the odd integers with $3 \leq m < 2025^2 = 4100625$, i.e. $3 \leq m \leq 4100623$. Their number is $\frac{4100623-3}{2} + 1 = 2050311$.

Total.

$$35 + 3 \cdot 2050311 = 35 + 6150933 = \boxed{6150968}.$$

There are **6150968** good polynomials.